

Sina Sajadmanesh

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SUMMARY

Senior AI Engineer specializing in multimodal foundation models, with a strong focus on translating research into high-performance, production-ready systems. Proven track record in building scalable model training pipelines and optimizing inference across heterogeneous hardware.

EDUCATION

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering	Aug 2023
Sharif University of Technology, MSc in Information Technology Engineering	Sept 2016
University of Isfahan, BSc in Computer Software Engineering	Feb 2014

EXPERIENCE

Sony AI , Senior AI Engineer – Zurich, Switzerland	Jan 2026 – present
- Optimized and deployed ML models on Qualcomm edge devices using QAIRT / SNPE / QNN, extending official support to modern LLMs and VLMs - Architected and maintained a modular ML acceleration and deployment framework supporting diverse hardware targets (Nvidia GPUs, Qualcomm NPUs, Apple Silicon) - Developed compact vision-language models supporting image/video understanding and localization tasks, achieving competitive performance with reduced computational requirements	
Sony AI , AI Engineer – Zurich, Switzerland	Oct 2023 – Dec 2025
- Designed a modular and flexible geometric optimization algorithm, enabling NeurIPS'25 spotlight paper - Led end-to-end optimization of Sony's Argus VFM inference stack, resulting in up to $61\times$ speedup across 17 vision tasks - Developed 7 computer vision models, including object detection, depth estimation, super-resolution, and OCR, achieving SOTA or competitive results - Architected and implemented a foundational MTL framework for Sony's Argus VFM, supporting 17 vision tasks, enabling CVPR'25 publication	
Idiap Research Institute , Research Assistant – Martigny, Switzerland	May 2019 – Aug 2023
- Worked on differentially private machine learning with graph neural networks 3 papers in top-tier conferences (CCS, USENIX Security, and WSDM) with 340+ citations 7 invited talks at top universities and research institutions, including Imperial College, UIC, and Twitter 6 open-source projects with 130+ stars on GitHub 1 short course taught on "Trustworthy Machine Learning" at Artificial Intelligence Doctoral Academy - Finalist in CSAW Applied Research Competition for the best paper award in computer security in Europe	
Brave Software , Research Intern – Remote	Mar 2022 – May 2022
Developed a novel privacy-preserving federated learning framework for neural bandit models under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite	

SKILLS AND EXPERTISE

Foundation & Generative Models: Vision & Multimodal Models (ViT, CLIP, LLaVA) · Large Language Models (LLaMA, GPT, Qwen) · Parameter-Efficient Fine-Tuning (LoRA) · Self-Supervised Learning (MAE, DINO)

Model Optimization & Deployment: TensorRT · ONNX · Qualcomm QNN / SNPE / QAIRT · Apple CoreML / MLX · Optimum · Quantization · Graph Optimization · Performance Profiling

Machine Learning Frameworks: PyTorch · Hugging Face (Transformers, PEFT, Accelerate, Datasets) · OpenMMLab (MMLCV, MMEngine, MMDetection)

Engineering & Tooling: Python · C++ · Bash · SQL · Data Processing (NumPy, Pandas, torchvision) · Environment & Packaging (uv, ruff) · CI/CD (Docker, GitHub Actions)

COMMUNITY AND PROFESSIONAL SERVICE

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), GNN User Group Meetup (2021), Twitter ML Seminar (2021)

Organizing / Program Committee: Privacy and Fairness in AI for Health (2023), PPAI (2024), WiseML (2023), PAIR2Struct (2022), DPML (2021)

Reviewer: NeurIPS (2026, 2024), CVPR (2026), TIFS (2025), IMWUT (2024), TDSC (2023), LoG (2023, 2022), AISTATS (2023), AIJ (2022), TBD (2021), TIST (2020), SNAM (2020), WWWJ (2018)

SELECTED PUBLICATIONS

StellA: Subspace Learning in Low-rank Adaptation using Stiefel Manifold Zhizhong Li, Sina Sajadmanesh, Jingtao Li, Lingjuan Lyu NeurIPS 2025 🌟 Spotlight (top 3%)	Dec 2025
Argus: A Compact and Versatile Foundation Model for Vision Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al. CVPR 2025	June 2025
ProGAP: Progressive Graph Neural Networks with Differential Privacy Guarantees Sina Sajadmanesh, Daniel Gatica-Perez WSDM 2024	Mar 2024
GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, Daniel Gatica-Perez USENIX Security 2023	Aug 2023
Locally Private Graph Neural Networks Sina Sajadmanesh, Daniel Gatica-Perez CCS 2021	Nov 2021
A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics Seyed Ali Osia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al. IEEE Internet of Things Journal	May 2020